

Correlation Uncertainty Module (SCUM) defines the governance conditions

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Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

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Governed Space

Correlation Uncertainty

Canonical Definition

Correlation Uncertainty Module (SCUM) defines the governance conditions under which uncertainty affecting the relationship between simulation and Operational Reality is identified, attributed where supportable, classified, bounded, propagated, documented, communicated, and preserved throughout Simulation-to-Reality Correlation.

SCUM governs uncertainty concerning:

how confidently a simulation can be said to correspond to reality within a defined object, condition, scenario, environment, time, comparison basis, and correlation scope.

Its purpose is not to eliminate uncertainty.

Its purpose is to prevent uncertainty from disappearing when simulation results are compared with reality, aligned against observations, or subsequently used as evidence.

Operational Role

SCUM is the fifth and final internal governance space of the Simulation-to-Reality Correlation Standard (SRCS).

The preceding modules establish:

SRCM — What simulation and reality observations may legitimately be compared?

SCCM — According to which criteria should correspondence be assessed?

SDMM — Where and how materially do simulation and reality differ?

SRAM — Which deviations require alignment, and how should that alignment occur?

SCUM asks:

How certain are we about the resulting Simulation-to-Reality Correlation, what remains unknown, and where does that uncertainty limit interpretation or downstream reliance?

The complete internal progression is:

Simulation-Reality Comparison

↓

Correlation Criteria

↓

Simulation-Reality Deviation

↓

Reality Alignment

↓

Correlation Uncertainty

Module Operational Space

SCUM governs: correlation uncertainty identification, uncertainty source identification, uncertainty classification, uncertainty attribution, uncertainty magnitude, uncertainty bounds, measurement uncertainty, observational uncertainty, representation uncertainty, model uncertainty, assumption-related uncertainty, environmental uncertainty, input uncertainty contribution, scenario uncertainty contribution, execution uncertainty contribution, temporal uncertainty, spatial uncertainty, behavioral uncertainty, interaction uncertainty, reality-reference uncertainty, comparison uncertainty, correlation-criteria uncertainty, deviation uncertainty, alignment uncertainty, residual uncertainty, epistemic

uncertainty, aleatory uncertainty, unknown uncertainty, uncertainty dependency, uncertainty interaction, uncertainty propagation, uncertainty accumulation, uncertainty reduction, irreducible uncertainty, condition-specific uncertainty, uncertainty thresholds, uncertainty materiality, uncertainty disclosure, uncertainty traceability, uncertainty status, and downstream uncertainty transfer.

Module Function

SCUM prevents a correlation statement such as:

“The simulation matches reality.”

from being treated as complete unless the uncertainty surrounding that statement is also visible.

A governed correlation statement should instead be capable of expressing:

what corresponds,

under which conditions,

against which reality basis,

with what observed deviations,

with what remaining uncertainty,

and:

where that uncertainty becomes material.

Correlation Is Not Certainty

SIMULOS® establishes:

Observed correlation does not eliminate uncertainty about the simulation-reality relationship.

Strong correspondence may coexist with uncertainty concerning: unobserved conditions, rare events, measurement limitations, model structure, environmental variability, future conditions, human behavior, autonomous behavior, interaction effects, or incomplete Operational Reality knowledge.

Correlation Uncertainty Object

Material Correlation Uncertainty should be capable of becoming a governed object.

Conceptually:

Correlation Uncertainty ID

↓

Affected Correlation Relationship



Uncertainty Source



Uncertainty Type



Affected Condition / Scenario



Magnitude / Bound where supportable



Dependencies



Materiality



Reducibility



Downstream Consequence



Uncertainty Status

SCUM should identify the origin of uncertainty where evidence permits.

Potential sources include: incomplete Operational Reality observations, measurement error, sensor limitations, sparse data, uncertain Reality Representation, uncertain assumptions, model simplification, unknown environmental conditions, input variability, scenario incompleteness, stochastic behavior, execution variability, incomplete comparison, uncertain correlation criteria, deviation measurement limitations, alignment effects, or unknown causes.

Unknown Source Must Remain Unknown

SIMULOS® establishes:

Uncertainty must not be assigned a cause merely because governance requires a

classification.

Where its source cannot be sufficiently established:

Uncertainty Source Undetermined

is a legitimate governed state.

Epistemic Uncertainty

Epistemic Uncertainty arises from incomplete knowledge.

Examples include: insufficient observations, incomplete model knowledge, uncertain parameter values, poorly understood interactions, limited environmental knowledge, and incomplete causal understanding.

Epistemic uncertainty may potentially be reduced through additional:

evidence, measurement, modeling, experimentation, or observation.

Aleatory Uncertainty

Aleatory Uncertainty arises from inherent variability or stochastic behavior in the represented reality.

It may not be eliminable simply by obtaining more information.

SCUM distinguishes:

uncertainty caused by what we do not know

from:

uncertainty caused by reality itself being variable.

Unknown Uncertainty

Some uncertainty cannot yet be meaningfully classified as epistemic or aleatory.

SCUM preserves:

Unknown / Unclassified Correlation Uncertainty

rather than forcing artificial certainty about the nature of uncertainty itself.

Reality-Reference Uncertainty

Operational Reality is not always perfectly observable.

Reality observations may themselves contain: measurement uncertainty, sampling limitations, missing states, incomplete telemetry, human reporting error,

sensor limitations, temporal gaps, or inaccessible conditions.

Therefore:

Reality Observation $\neq$ Perfect Reality Knowledge.

Simulation Uncertainty and Reality Uncertainty Must Remain Distinguishable

A simulation-reality difference may arise from:

Simulation Uncertainty

Reality Observation Uncertainty

or:

Both.

SCUM should preserve this distinction where possible.

Comparison Uncertainty

Even where both simulation and reality observations are individually strong, uncertainty may exist in whether they are sufficiently comparable.

Possible causes include: different environmental conditions, different timing, different configurations, different measurement resolution, different object states, or incomplete contextual equivalence.

Criteria Uncertainty

Correlation Criteria may themselves contain uncertainty.

For example, an organization may not know whether:

$\pm 2\%$

or:

$\pm 5\%$

represents a materially sufficient correspondence boundary for a particular operational purpose.

SCUM preserves uncertainty in the criterion rather than hiding it inside the correlation result.

Deviation Uncertainty

An observed deviation may have uncertainty around: magnitude, direction, frequency, localization, cause, propagation, or materiality.

Therefore:

Deviation Record

and:

Deviation Uncertainty

should remain distinguishable.

Alignment Uncertainty

A Reality Alignment action may improve observed correlation while creating uncertainty about: generalization, unobserved scenarios, model interactions, new parameter regions, or unintended downstream effects.

Alignment therefore does not automatically reduce total uncertainty.

Residual Correlation Uncertainty

After comparison, deviation assessment, and alignment, uncertainty may remain.

This becomes:

Residual Correlation Uncertainty.

Residual uncertainty should be explicitly preserved rather than implicitly converted into confidence.

Uncertainty Magnitude

Where evidence permits, uncertainty may be expressed quantitatively through: confidence intervals, probability distributions, error ranges, tolerance intervals, sensitivity ranges, variance, credible intervals, or other appropriate methods.

Where numerical quantification is not supportable, structured qualitative classification may be used.

No False Precision Principle

SIMULOS® establishes:

Correlation uncertainty must not be expressed more precisely than the available evidence permits.

A numerical percentage can create misleading authority where the underlying evidence cannot support that precision.

Qualitative Uncertainty Classification

Where quantitative measurement is inappropriate, implementation may use classifications such as:

Low

Moderate

High

Critical

or:

Undetermined.

These classifications must remain bounded to the applicable correlation context.

Uncertainty Bounds

Where possible, SCUM should establish the range within which the correlation relationship is reasonably supported.

For example:

Strong Correlation — Nominal Conditions

Moderate Uncertainty — High Temperature

High Uncertainty — Extreme Combined Conditions

Unknown — Unobserved Failure Configuration

This is more informative than a single global confidence score.

Condition-Specific Uncertainty

Correlation uncertainty may vary significantly across conditions.

Therefore:

Correlation Confidence is a surface, not necessarily a single number.

A simulation may be highly correlated with reality in one operational region and poorly understood in another.

Scenario-Specific Uncertainty

Each materially relevant scenario may possess a different uncertainty profile.

This is particularly important for: edge cases, stress conditions, rare events, novel environments, and combinations never directly observed in Operational Reality.

Temporal Uncertainty

A simulation may correspond well with current reality while uncertainty increases over time because: the environment changes, system behavior evolves, components degrade, software changes, populations change, or Operational

Reality drifts.

SCUM therefore allows correlation uncertainty to be time-dependent.

Spatial Uncertainty

Correlation may be strong in observed geographic or physical regions while remaining uncertain elsewhere.

This matters for: autonomous vehicles, robotics, climate simulation, aerospace, infrastructure, and geographically distributed systems.

Behavioral Uncertainty

Behavioral uncertainty concerns uncertainty about whether simulated actors or systems behave sufficiently like their real counterparts.

This can be particularly material for: humans, AI agents, autonomous systems, multi-agent systems, adaptive systems, and adversarial environments.

Interaction Uncertainty

Even where individual components are well represented, their interactions may remain uncertain.

SCUM therefore governs uncertainty arising from:

component × component

human × system

agent × agent

system × environment

and other interaction relationships.

Uncertainty Dependency

Uncertainties may depend upon one another.

For example:

Environmental Uncertainty

↓

Sensor Uncertainty

↓

Perception Uncertainty

↓

Behavioral Uncertainty

↓

Trajectory Correlation Uncertainty

SCUM should preserve material dependencies where identifiable.

Uncertainty Interaction

Multiple individually modest uncertainties may interact to produce a materially larger uncertainty.

For example:

Sensor Noise Weather Variability Timing Uncertainty

may create:

High Autonomous Behavior Correlation Uncertainty.

Uncertainty Propagation

Correlation uncertainty may propagate downstream through the simulation lifecycle and into later governance architectures.

Conceptually:

Correlation Uncertainty

↓

Simulation Output Interpretation

↓

Validation Evidence

↓

Validation Determination

↓

Operational Reliance

SCUM therefore requires material uncertainty to remain transferable rather than disappearing at the SIMULOS® boundary.

No Uncertainty Laundering Principle

SIMULOS® establishes:

Uncertainty identified within Simulation-to-Reality Correlation must not disappear merely because the simulation output enters another governance process.

If VALIDOS® uses the simulation as validation evidence, the relevant uncertainty must travel with it.

Uncertainty Accumulation

Where multiple uncertain relationships contribute to one correlation conclusion, SCUM should determine whether uncertainty accumulates materially.

A simulation should not claim high overall certainty merely because each individual uncertainty appears small when assessed separately.

Uncertainty Reduction

Where uncertainty is reducible, SCUM may identify actions such as: additional Operational Reality observations, additional measurements, broader scenario execution, targeted experiments, higher-fidelity representation, model refinement, parameter reassessment, additional comparison, or repeated execution.

Reducible vs Irreducible Uncertainty

SCUM distinguishes:

Reducible Uncertainty

from:

Currently Irreducible Uncertainty.

This matters because governance responses differ.

Reducible uncertainty may create an evidence or simulation improvement requirement.

Irreducible uncertainty may instead require:

bounded interpretation, explicit conditions, or restricted reliance.

Uncertainty Reduction Has a Cost

Reducing uncertainty may require: computation, testing, physical experiments, additional missions, new sensors, more data, time, or operational exposure.

SIMULOS® does not require uncertainty to be reduced indefinitely.

The appropriate reduction depth depends upon the simulation purpose and

downstream consequence.

Material Uncertainty

Uncertainty becomes material when it can materially affect: interpretation, correlation classification, simulation output meaning, validation, safety assessment, operational decision-making, autonomous-system behavior, mission planning, or reliance.

Uncertainty Threshold

Where appropriate, implementations may establish thresholds beyond which:

additional evidence,

additional simulation,

alignment,

validation restriction,

or:

reliance limitation

is required.

SIMULOS® does not prescribe one universal uncertainty threshold.

Correlation Confidence

An implementation may derive a bounded Correlation Confidence representation from the governed uncertainty basis.

However:

Correlation Confidence must never conceal the conditions, scope, or unresolved uncertainty upon which it depends. Correlation Confidence $\neq$ Probability of Reality

A confidence representation must not imply that the simulation itself has become reality or that all future real-world behavior has been probabilistically captured.

Correlation Uncertainty Status

Possible implementation states may include:

Sufficiently Bounded

Conditionally Bounded

Material Uncertainty Present

High Uncertainty

Critical Uncertainty

or:

Uncertainty Undetermined.

Correlation Uncertainty Record

A governed record may include:

Correlation Uncertainty ID

Simulation Object / Version

Reality Reference

Comparison Basis

Affected Scenario / Condition

Uncertainty Source

Uncertainty Type

Magnitude / Range

Dependencies

Propagation Path

Materiality

Reducibility

Reduction Requirement

Residual Uncertainty

Downstream Limitation

Status

Evidence Basis

Timestamp / Version Context Uncertainty

Across Versions

SCUM should support comparison of uncertainty across simulation versions.

For example:

Simulation v1 → High uncertainty in thermal extremes

Simulation v2 → Moderate uncertainty

Simulation v3 → Low uncertainty within tested range / Unknown beyond tested range

This creates a more meaningful improvement record than simply declaring newer versions “more accurate.” Uncertainty Trend

Repeated simulation-reality correlation may reveal:

Decreasing Uncertainty

Stable Uncertainty

Increasing Uncertainty

Migrating Uncertainty

or:

Newly Emergent Uncertainty.

These trends may indicate where simulation governance should re-enter upstream spaces.

Correlation Coverage and Uncertainty

High correlation inside a narrow observed region does not establish low uncertainty outside that region.

Therefore:

Observed Correlation Coverage must remain distinguishable from Unobserved Correlation Uncertainty.

Unknown Regions

Where no sufficient Operational Reality basis exists, SCUM should permit explicit classification such as:

Correlation Unknown — Insufficient Reality Observation.

This is especially important where simulation exists precisely because direct real-world testing is difficult or impossible.

Novel Conditions

A simulation may be used to explore conditions that have never occurred or cannot safely be reproduced.

SCUM does not prohibit this.

Instead, it requires the distinction between:

Empirically Correlated Region

and:

Simulation-Only / Uncorrelated Region.

Extrapolation Uncertainty

Where simulation extends beyond observed reality, SCUM should identify:

Extrapolation Uncertainty.

The farther the simulation moves beyond the governed correlation basis, the more carefully its correspondence claims should be bounded.

No Correlation Transfer by Similarity

SIMULOS® establishes:

Correlation established for one condition, environment, configuration, or system version must not automatically transfer to another merely because the two appear similar.

Transfer requires a governed basis.

Cross-Version Reality Correlation

A new system or simulation version may inherit some correlation evidence from a previous version only where the relevant relationship remains materially comparable.

Otherwise new correlation assessment is required.

Correlation Uncertainty and Decision Consequence

The same uncertainty level may be acceptable for:

early design exploration

but unacceptable for:

safety-critical deployment authorization.

SCUM therefore preserves downstream consequence context without itself making the final reliance decision.

Uncertainty Communication

Material uncertainty should be communicated in a form usable by: engineers, decision-makers, validators, auditors, regulators, autonomous systems, and downstream AI systems.

The communication should preserve both:

what is known

and:

what remains uncertain.

Machine-Readable Uncertainty

For AI-readable implementations, uncertainty should be capable of being represented structurally.

Conceptually:

Correlation Scope

Correlation Status

Uncertainty Type

Uncertainty Bound

Applicable Conditions

Excluded Conditions

Known Deviations

Unknown Regions

Downstream Restrictions

This allows downstream systems to consume uncertainty rather than merely consuming a correlation score.

Relationship to SRCM

SRCM determines whether simulation and reality observations are sufficiently comparable.

SCUM preserves uncertainty remaining in that comparison basis.

Relationship to SCCM

SCCM establishes the criteria for judging correspondence.

SCUM governs uncertainty affecting the interpretation and application of those criteria.

Relationship to SDMM

SDMM governs observed Simulation-Reality Deviations.

SCUM governs uncertainty concerning those deviations and the broader correlation relationship.

Therefore:

Deviation $\neq$ Uncertainty.

A deviation is an observed difference.

Uncertainty concerns how confidently that difference and the resulting correlation relationship are understood.

Relationship to SRAM

SRAM governs Reality Alignment.

SCUM determines what uncertainty remains before, during, and after alignment.

Therefore:

Alignment may reduce deviation without eliminating uncertainty.

Relationship to ORA™

ORA™ governs Operational Reality.

SCUM does not assign certainty to Operational Reality itself.

It governs how uncertainty in the available reality basis affects the simulation-to-reality relationship.

Relationship to VALIDOS®

VALIDOS® may use simulation and Simulation-to-Reality Correlation as part of a Validation Evidence basis.

SCUM ensures that material correlation uncertainty remains attached to that evidence.

Therefore:

VALIDOS® must not receive simulation correlation as stronger evidence than SIMULOS® has actually established.

Relationship to INTEGROS®

INTEGROS® may govern integrity properties of: observations, measurements, datasets, records, and uncertainty artifacts.

SCUM governs the simulation-specific uncertainty relationship.

Relationship to SOGS

SOGS governs what simulation outputs are and what may legitimately be inferred from them.

SCUM provides a correlation-uncertainty basis that may materially constrain those inferences where outputs are interpreted as representations of reality.

Relationship to Operational Reliance

SCUM does not itself authorize operational reliance.

It provides the uncertainty basis required by downstream governance to determine whether reliance is legitimate.

Therefore:

Correlation Uncertainty → Reliance Input

not:

Correlation Uncertainty → Reliance Authorization.

Autonomous Systems Application

An autonomous system may show excellent simulation-reality correlation across millions of nominal simulation events.

However, limited real-world observations may exist for: rare pedestrian behavior, unusual weather combinations, degraded sensors, adversarial interactions, infrastructure failures, multi-agent conflicts, or emergency recovery.

SCUM prevents nominal correlation from being generalized automatically into low uncertainty across the entire autonomous operating domain.

A machine-readable Correlation Uncertainty Map can identify where the system's simulation evidence is strong and where real-world correspondence remains uncertain.

Space and Mission Simulation Application

Space agencies frequently rely on simulation precisely where direct reality evidence is sparse.

A spacecraft or planetary-landing simulation may possess strong correlation evidence for: subsystem tests, terrestrial hardware testing, known orbital conditions, previous mission telemetry, and observed environmental parameters.

But uncertainty may remain high for: never-observed surface interactions, extreme radiation combinations, long-duration degradation, unknown atmospheric behavior, rare subsystem combinations, autonomous recovery, or first-of-kind mission conditions.

SCUM allows those regions to remain explicitly:

Simulation-Supported but Reality-Correlation Uncertain

rather than silently inheriting the confidence of well-tested regions.

This can reveal exactly where additional testing, observation, instrumentation, scenario coverage, or conservative mission constraints provide the greatest value.

Minimum Implementation Framework

1. Establish the Correlation Basis

Identify the simulation, Operational Reality reference, comparison basis, criteria, deviations, and alignment state. 2. Identify Material Uncertainty

Determine where the correlation relationship remains uncertain. 3. Identify the Uncertainty Source

Attribute uncertainty where evidence permits.

Otherwise record:

Source Undetermined. 4. Classify the Uncertainty

Determine whether uncertainty is: epistemic, aleatory, mixed, or unknown.

5. Localize the Uncertainty

Identify the affected: condition, scenario, component, environment, behavior, interaction, time, or output.

6. Bound or Quantify Where Supportable

Use numerical or qualitative bounds appropriate to the evidence. 7. Map Dependencies and Interaction

Determine whether uncertainties depend upon or amplify one another. 8. Assess Propagation

Identify where uncertainty can affect downstream interpretation or governance.

9. Determine Materiality

Assess whether the uncertainty materially affects the intended simulation purpose or downstream use. 10. Determine Reducibility

Identify whether additional evidence or simulation activity can materially reduce the uncertainty. 11. Preserve Residual Uncertainty

Record what remains unresolved after available reduction or alignment actions.

12. Establish Correlation Uncertainty Status

Assign a bounded status appropriate to the applicable correlation scope. 13. Transfer Material Uncertainty Downstream

Ensure that simulation outputs, validation processes, audits, or reliance decisions receive the relevant uncertainty information. 14. Preserve Traceability

Maintain the relationship:

Reality Basis → Comparison → Criteria → Deviation → Alignment → Uncertainty → Downstream Use.

Governance Outputs

SCUM may produce: Correlation Uncertainty ID, Correlation Uncertainty Record, Uncertainty Source Record, Uncertainty Classification, Epistemic Uncertainty Record, Aleatory Uncertainty Record, Unknown Uncertainty Record, Reality-Reference Uncertainty Record, Comparison Uncertainty Record, Criteria Uncertainty Record, Deviation Uncertainty Record, Alignment Uncertainty Record, Residual Correlation Uncertainty Record, Quantitative Uncertainty Bound, Qualitative Uncertainty Classification, Condition-Specific Uncertainty Map, Scenario-Specific Uncertainty Map, Temporal Uncertainty Record, Spatial Uncertainty Record, Behavioral Uncertainty Record, Interaction Uncertainty Record, Uncertainty Dependency Map, Uncertainty Propagation Map, Uncertainty Accumulation Assessment, Uncertainty Reduction Requirement, Reducibility Assessment, Extrapolation Uncertainty Record, Unknown Correlation Region Record, Correlation Uncertainty Status, Correlation Confidence Record, Downstream Uncertainty Transfer Record, and Governed Simulation-to-Reality Correlation Uncertainty Basis.

Use Case 1 — Autonomous Driving Beyond Tested Conditions

An autonomous driving simulation demonstrates strong correlation with Operational Reality across extensive urban driving tests.

However, only limited real-world evidence exists for:

heavy snow + partially blocked sensors + unusual pedestrian movement + low visibility.

SCUM prevents the strong nominal correlation from automatically extending into this region.

The system records:

Nominal Urban Conditions → Correlation Uncertainty Low

Heavy Snow → Correlation Uncertainty Moderate

Heavy Snow + Sensor Degradation + Rare Pedestrian Interaction → Correlation Uncertainty High

This uncertainty can then travel with the simulation evidence into validation

and operational decision-making. Use Case 2 — First-of-Kind Space Mission

A space agency uses simulation to assess an autonomous landing sequence for a destination where direct Operational Reality observations are necessarily limited.

The simulation correlates strongly with: terrestrial hardware testing, propulsion tests, sensor testing, and comparable prior mission data.

However, the exact combination of:

surface properties + environmental conditions + autonomous response + landing dynamics

has never been directly observed.

SCUM prevents correlation from known regions from being silently transferred into the unknown mission region.

The resulting correlation basis may state:

Subsystem Correlation — Strong

Known Environmental Range — Sufficiently Bounded

Exact Landing Environment — Material Epistemic Uncertainty

Combined First-of-Kind Landing Interaction — Correlation Unknown / Simulation-Supported

This gives engineers and decision-makers a governed map of where additional evidence can reduce uncertainty and where uncertainty must instead remain an explicit mission constraint.

Architectural Position

SCUM is the fifth and final module of the Simulation-to-Reality Correlation Standard (SRCS).

The complete SRCS progression is:

SRCM — Simulation-Reality Comparison

↓

SCCM — Simulation Correlation Criteria

↓

SDMM — Simulation-Reality Deviation

↓

SRAM — Reality Alignment

↓

SCUM — Correlation Uncertainty

Together, these five modules answer:

What can be compared?

↓

According to what criteria?

↓

Where does simulation differ from reality?

↓

What should be aligned?

↓

How certain are we about the resulting relationship?

This closes the governed Simulation-to-Reality Correlation space without converting correlation into validation or simulation into reality.

Foundational Principle

Simulation similarity is not reality equivalence, and correlation confidence must never exceed the reality evidence, comparison basis, observed conditions, and unresolved uncertainty that actually support it.

Canonical Closing Statement

Correlation Uncertainty Module (SCUM) completes the Simulation-to-Reality Correlation Standard by governing what remains uncertain after simulation and Operational Reality have been compared, deviations identified, and justified alignment performed. By distinguishing simulation uncertainty from reality-reference uncertainty, epistemic uncertainty from aleatory variability, observed correlation from unobserved regions, deviation from uncertainty, alignment from certainty, and empirical correspondence from extrapolation, SCUM prevents strong correlation in known conditions from becoming an unsupported claim about reality as a whole. Through uncertainty identification, classification, bounding, dependency mapping, propagation, reduction, residual-uncertainty preservation, machine-readable disclosure, and downstream transfer, SCUM ensures that every Simulation-to-Reality Correlation remains only as certain, broad, and operationally meaningful as the evidence actually permits.

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